Here are **production-focused Kubernetes DaemonSet interview questions and answers**, tailored to real-world scenarios and troubleshooting insights:

🔹 1. What is a DaemonSet, and how is it used in production?

Answer:
A DaemonSet ensures that a specific pod runs on **every (or selected) node** in the cluster. It's commonly used in production for deploying:

- Log collectors (e.g., Fluentd)
- Monitoring agents (e.g., Prometheus Node Exporter)
- Security tools (e.g., Falco, antivirus scanners)
- System daemons (e.g., disk cleanup, node-level services)

🔹 2. What happens when a new node is added to a cluster with a DaemonSet?

Answer:
The Kubernetes controller automatically schedules the DaemonSet pod on the **new node**, without requiring a manual update. This ensures consistent functionality across the entire node fleet.

🔹 3. Can you run multiple pods per node with a DaemonSet?

Answer:
Not directly. DaemonSet is designed to run **one pod per node**. If multiple are needed, consider:

- Using multiple DaemonSets with different labels.
- Having a pod spawn sidecar containers or manage subprocesses internally.

🔹 4. How do you restrict a DaemonSet to run only on certain nodes?

Answer:
Use **node selectors**, **node affinity**, or **taints and tolerations**. For example:

```yaml
spec:
  template:
    spec:
      nodeSelector:
        role: logging
```

or with affinity:

```yaml
affinity:
  nodeAffinity:
    requiredDuringSchedulingIgnoredDuringExecution:
      nodeSelectorTerms:
      - matchExpressions:
        - key: role
          operator: In
          values:
          - monitoring
```

🔹 5. How do you update a DaemonSet in production without downtime?

Answer:
Use a **rolling update strategy**:

```
updateStrategy:
  type: RollingUpdate
  rollingUpdate:
    maxUnavailable: 1
```

This ensures that only one pod is unavailable at a time, minimizing impact.

🔹 6. How do you debug a DaemonSet pod that's not being scheduled?

Answer:

- Check pod events: `kubectl describe pod <name>`
- Common issues:
    - NodeSelector mismatch
    - Missing tolerations for tainted nodes
    - Image pull errors
    - SecurityContext or permissions issues

🔹 7. Can DaemonSets be used on master/control plane nodes?

Answer:
Yes, but only if:

- You **tolerate** the node-role.kubernetes.io/master taint
- You target them specifically (via label or affinity)

```
tolerations:
- key: "node-role.kubernetes.io/master"
  effect: "NoSchedule"
```

🔹 8. What's the difference between a Deployment and a DaemonSet in production?

Feature	Deployment	DaemonSet
Purpose	N replicas anywhere	1 per (selected) node
Scheduling	Cluster-wide	Node-specific
Use case	App services	Agents, daemons, logs
Replica control	User-defined	Node-defined
